

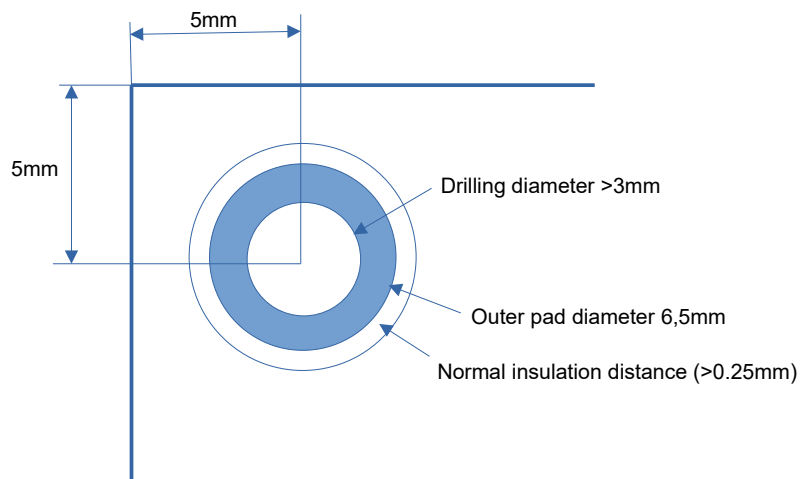
Summary of the requirements for PCB production on milling machines

A circuit board, especially for laboratory tests/experiments, should have:

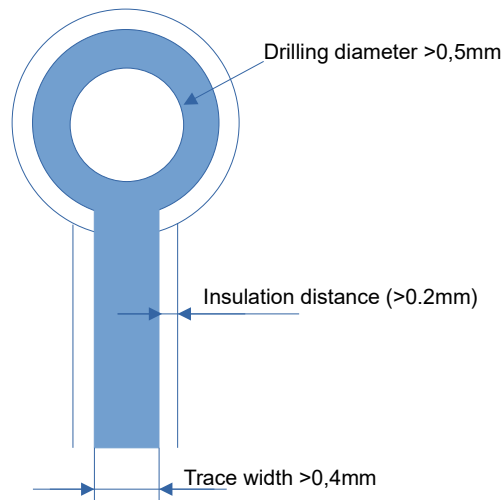
- Connection sockets for power supply via 4mm laboratory plug on the top of the circuit board
- Jumpers must be labelled (top layer or top overlay)
- Leave sufficient space to pull the jumpers (operability)
- Test points for:
 - important signals that should then be part of the elaboration
 - Provide sufficient GND taps, especially if several signal curves are to be displayed simultaneously
- Labelling the test points
- Blocking capacitors
 - 10uF wired on top side (cup capacitors)
 - 100nF on bottom/top side near the critical components (SMD)
- Version notes/creation date on circuit boards
- Check dimensions
- Font height 80 mil
- Font width 15 mil (line width still easy to mill)

Mechanical requirements, dimensions:

- Provide fastening options (here for M3):
 - Spacer bolts can be used to set the boards down Possibly provide a certain minimum length to make the boards stackable
 - 3mm (120mil) hole at each corner
 - It is best to place a pad for this
 - Pad diameter 6.5 mm (250mil) electrically insulated from the rest of the board
 - The hole should have a distance of approx. 5mm (200mil) to the edge in each case



- Requirements for tracks and drill holes:
 - Insulation absolute minimum distance $>0.2\text{ mm}$ (better 0.3 mm)
 - Absolut smallest track width approx. 0.2 mm (better 0.4 mm otherwise problematic when setting the depth of the cutter)
 - Smallest drill diameter $>0.5\text{ mm}$
 - Minimum distance to neighbouring conductor tracks equals insulation distance



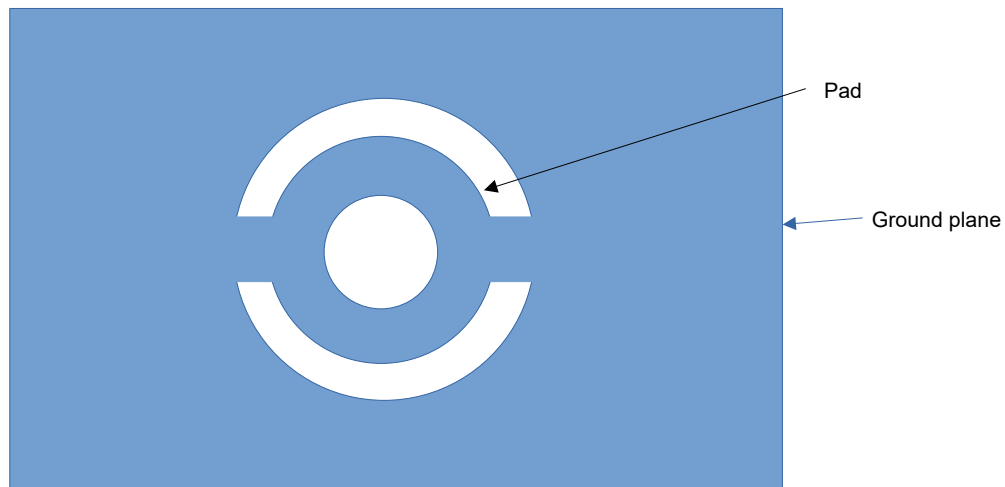
- Current carrying capacity according to the following table

Example:

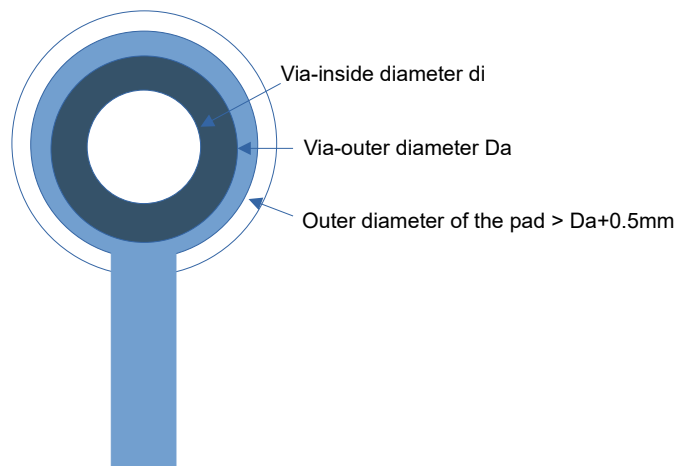
A track 1.0 mm wide on $35\text{ }\mu\text{m}$ copper will heat to approx. $20\text{ }^{\circ}\text{K}$ above room temperature when a 3 amp current is applied. These values may vary greatly depending on the structure of the board of the board and on the environment and air circulation.

CU Schicht Stärke	Leiterbahn breite	Max. Strom in Abhängigkeit zur Temperaturerhöhung (delta Kelvin)				
Copper thickness on layer	Track-wide	Max. Current depending on tepmerature increase (delta Kelvin)				
		10 °K	20°K	30°K	45°K	60°C
35µm 1 oz	0,25mm	0,5A	0,8A	1,0A	1,3A	1,6A
	0,50mm	1,0A	1,6A	2,0A	2,5A	3,0A
	1,00mm	2,2A	3,0A	3,6A	4,2A	4,8A
	1,50mm	3,0A	3,8A	4,6A	5,3A	6,5A
	2,00mm	3,8A	5,0A	6,5A	7,5A	8,5A
	3,00mm	4,5A	6,5A	8,0A	9,5A	11,0A
	4,00mm	6,0A	8,5A	10,0A	12,0A	13,5A
	5,00mm	7,0A	10,0A	12,0A	14,5A	16,0A
	6,00mm	7,5A	11,0A	14,0A	16,0A	18,0A
	8,00mm	9,0A	14,0A	17,0A	20,0A	22,5A
	10,00mm	10,0A	16,0A	20,0A	23,0A	26,0A
70µm 2 oz	0,25mm	1,0A	1,6A	2,0A	2,5A	3,0A
	0,50mm	2,0A	2,8A	3,5A	4,0A	4,5A
	1,00mm	3,5A	4,7A	5,8A	6,8A	8,0A
	1,50mm	4,5A	6,2A	7,5A	9,0A	10,5A
	2,00mm	6,0A	8,5A	10,0A	12,0A	13,5A
	3,00mm	7,5A	11,0A	14,0A	16,0A	18,0A
	4,00mm	9,0A	13,5A	17,0A	19,0A	22,0A
	5,00mm	10,0A	15,0A	19,0A	23,0A	25,0A
	6,00mm	11,0A	18,0A	22,0A	26,0A	28,0A

- Requirement for connection to large copper surfaces (self-explanatory):



- Requirements for vias:



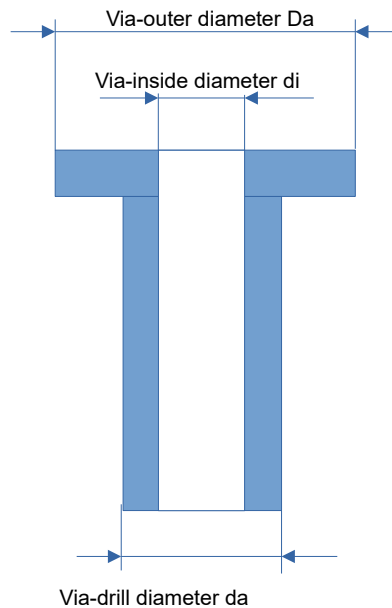
As a rule, we do not produce vias galvanically. It is possible to place vias manually using hollow rivets. Each individual rivet set in this way must then be pressed again.

In practice, it often makes more sense to drill smaller holes, i.e. smaller vias. The top and bottom sides are then connected to each other by soldering shortened pieces of wire.

It should be noted that the contacts of wired components must be accessible on both sides.

In the case of manufacturing PCBs with milling technique, a via must be placed at this point. The inner diameter of the via must then be larger than the wire diameter.

With some wired components (resistors, for example), the connecting wire itself can be used as a via. This is not possible with cup electrolytic capacitors, as the wire is not accessible from above (on the cup side) and therefore cannot be soldered.



di	da	Da	Material
0,4	0,6	1,0	Silver
0,6	0,8	1,2	Silver
0,6	0,8	1,4	Copper
0,7	1,0	1,4	Silver
0,8	1,0	1,6	Copper
1,0	1,4	2,3	Copper

(Via types; dimensions in mm; crossed-out types are not in stock, whether pressing tools are available must always be checked beforehand; vias marked in green are preferably used).

The manufacturer's specifications/limits must be observed for externally manufactured PCBs. This applies to all of the above considerations.

- Data required for milling machines (CAM):
 - Drilling plated (drill holes)
 - Copper top (conductor tracks on the top)
 - Copper bottom (conductor tracks on the underside)
 - Cutting Outside (outer edge of the circuit board)

The data must be available as Gerber data. If possible, the data should be exported as such:

- In inch format
- Data format 2:3 (number of digits before and after the decimal point)
- Leading and trailing zeros preserved
- Reference to relative zero point

In the case of deviations, data processing may be somewhat more time-consuming because the data formats have to be "tried out".

And finally allways use filled layers where possible this lowers board manufacturing costs.